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Illuminating T cell-dendritic cell interactions in vivo by FlAsHing antigens

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Abstract

Delineating the complex network of interactions between antigen-specific T cells and antigen presenting cells (APCs) is crucial for effective precision therapies against cancer, chronic infections, and autoimmunity. However, the existing arsenal for examining antigen-specific T cell interactions is restricted to a select few antigen-T cell receptor pairs, with limited in situ utility. This lack of versatility is largely due to the disruptive effects of reagents on the immune synapse, which hinder real-time monitoring of antigen-specific interactions. To address this limitation, we have developed a novel and versatile immune monitoring strategy by adding a short cysteine-rich tag to antigenic peptides that emits fluorescence upon binding to thiol-reactive biarsenical hairpin compounds. Our findings demonstrate the specificity and durability of the novel antigen-targeting probes during dynamic immune monitoring in vitro and in vivo. This strategy opens new avenues for biological validation of T-cell receptors with newly identified epitopes by revealing the behavior of previously unrecognized antigen-receptor pairs, expanding our understanding of T cell responses.

eLife assessment

This is an **important** study that develops a method to fluorescently label peptide MHC complexes on live dendritic cells to enable detection of antigen specific T cells in polyclonal populations. **Solid** evidence that this can be used to effectively identify antigen specific T cells in vitro and in vivo is provided for one model antigen systems (Ova-OTII). The approach has exciting potential as prior single step methods with directly conjugated single peptides have generally failed due to high background. Thus, this approach potentially moves the state of the art forward, but further work is needed to realise and determine the limits and ultimate utility of the approach.

Introduction

The interactions between T cells and APCs form the cornerstone of adaptive immunity. Within the intricate milieu of lymphoid tissue, T cell priming stands out as a discrete and precise process, occurring amidst the multitude of dynamic cellular interactions. Multimerized peptide-major histocompatibility complex (pMHC) labeling has been the gold standard tool for capturing antigen-specific T cells from a polyclonal repertoire [1]. However, such multimers, most commonly tetramers, have limitations in monitoring dynamic interactions of antigen-specific T cells in real-time due to their blocking effect on the immune synapse. Likewise, currently available T cell receptor like (TCRL), also known as peptide-in-groove (PIG), antibodies that detect pMHC complexes are known to disrupt the immune synapse, making them unsuitable for real-time monitoring of antigen-specific T cell interactions [2, 3]. Although alternative approaches for labeling antigenic peptides with conventional tags such as fluorescent proteins, fluorophores, and biotin-streptavidin have been used, these tags have limitations such as size constraints, membrane impermeability, and photophysically unstable nature, which restrict their use in applications for dynamic monitoring [4-7].

To overcome these drawbacks, we have developed a highly specific and minimally disruptive labeling strategy for studying antigen-specific T cell-APC interactions. Our approach involves introducing a six-amino acid tetracysteine tag (CCPGCC) on antigenic peptides and inducing fluorescence using membrane permeable thiol-reactive Arsenical Hairpin (AsH) probes [5, 8]. We have engineered the $OVA_{(323\text{-}339)}$ peptide by placing the tetracysteine tag at either the amino or the carboxy terminus, away from the TCR and MHC binding sites, to avoid disrupting T cell- APC synapse in vitro and in vivo [9]. Owing to these modifications, our labeling strategy preserves the T cell priming ability and allows for the detection of the fluorescent signal on the surface of APCs in a MHCII-dependent manner. Using flow cytometry and live microscopy, we found that the labeled peptide is readily taken up by the cognate $CD4^+$ T cells with precise specificity in vivo, as previously shown for pMHC complexes taken up from the immune synapse via processes such as trogocytosis and transendocytosis [10-14]. Furthermore, we have created new variants of the tetracysteine-tagged $OVA_{(323\text{-}339)}$ peptide with altered TCR-binding sites and have shown that the AsH probe labeling can track the TCR mediated uptake of nuanced pMHC complexes. This indicates that our labeling strategy is highly robust and forges a novel path for investigating immune responses against altered antigens such as those derived from self, microbiota and tumor microenvironment. Overall, AsH probe mediated immune monitoring offers significant advantages over existing methods, making it a versatile tool for in vitro and in vivo applications including live imaging of the T cell synapse within intact tissue architecture. This novel approach will further our understanding of T cell-mediated immune responses by also complementing tetramers for sorting antigen-specific T cells with precision and extending it to newly discovered epitopes.

Results

Signal intensity is determined by the sequence and placement of tetracysteine tag

We first generated $OVA_{323\text{-}339}$ variants by placing a tetracysteine tag “CCPGCC” at amino or carboxy termini, either consecutively or via a flexible ε-amino-caproic acid (ε-ACA) linker to prevent endosomal-lysosomal enzymatic digestion to the tag. This yielded four variant peptides named as C- (carboxy terminus without linker), N- (amino terminus without linker), NACA- (amino terminus with ε-ACA) and CACA- (carboxy terminus with ε-ACA) (**Fig 1a**). We used these variants to load mature C57BL/6 $CD11c^+$ splenic DCs from C57BL/6 mice and induced fluorescence by treating cells with Fluorescein-AsH (FlAsH) probe. Flow cytometry analysis showed that CACA-peptide combination but not the other tested variants induced a distinguished fluorescent signal (**Fig 1b**). Cysteine is naturally oxidation prone and this may reduce FlAsH binding to tetracysteine tag. We addressed this problem by pre-treating OVA_{CACA} with tris-2-carboxyethylphosphine (TCEP), an irreversible reducing agent that blocks cystine disulfide bridge formation and elicited a consistent FlAsH signal (**Fig. 1c**). Moreover, we used another AsH probe excitable by 561 nm laser, called ReAsH, to occupy non-specific cysteine residues that would otherwise attract FlAsH [15]. Pre-incubation of DCs with ReAsH improved signal to noise ratio significantly (**Fig 1d**). We have created additional OVA variants to enhance signal intensity. These variants include one with two tetracysteine residues and another with a longer tetracysteine tag that withstands the British anti-Lewisite (BAL) wash, a crucial step to reduce background labeling by removing excess AsH probe [16]. However, new variants provided FlAsH intensity comparable to the original CACA variant, indicating no additional benefit for longer tags (**Fig 1e**).

Tetracysteine tag does not alter the immunogenicity of peptide

Next, we sought to compare the CD4$^+$ T cell priming ability of four tetracysteine-tagged OVA$_{323\text{-}339}$ variants and that of native peptide in vitro. To do so, naïve CD4$^+$ T cells were isolated from OVA$_{323\text{-}339}$ specific TCR transgenic OTII mice, labelled with a proliferation dye and cocultured with mature CD11c$^+$ DCs that were pre-loaded with equimolar amount peptides for three days. Proliferation and activation status of OTII T cells were not altered by the presence or placement of the tag (**Fig 2a-d**). Subsequently, DCs were pulsed with OVA$_{CACA}$ vs. native peptide, treated with FlAsH and cocultured with fluorescently labeled OTII T cells for live confocal microscopy imaging. Within the first hours of coculture, OTII T cells established stable contact with OVA$_{CACA}$-pulsed DCs and FlAsH-labeled peptide gradually concentrated at the contact site (**Fig 2e**). At 24 hours after treatment, OTII T cells in the peptide-treated groups exhibited increased volume and a shift towards a less spherical shape, indicative of T cell blasting (**Fig 2f**). Furthermore, the interaction of T cells with OVA$_{CACA}$ and native OVA$_{323\text{-}339}$ pulsed DCs resulted in a comparable decrease in T cell motility, suggesting that the tag had no impact on their dynamics in vitro (**Fig 2g**).

Next, we performed a series of adoptive transfers to measure immunogenicity and signal persistence of tetracysteine-tagged peptides in vivo using two-photon microscopy and flow cytometry. Splenic DCs from β-Actin DsRed mice were treated with ReAsH to prime them for FlAsH labeling. Because DsRed and ReAsH do not get transferred to T cells during immune synapse, their overlapping fluorescence emission do not pose any problem for tracking the DCs. Next, DCs were pulsed with a peptide mix comprised of equimolar amounts of OVA-Biotin and OVA$_{CACA}$, where OVA-Biotin served as a positive control for flow cytometry. This was followed by optimized FlAsH treatments as indicated in **Fig.1** and adoptive transfer into C57BL/6 recipients via footpad injection. Naïve CD4$^+$ T cells were isolated from OTII and WT C57BL/6 spleens, labelled with distinct cell-tracker dyes, mixed 1:1 and transferred into recipients via retroorbital i.v. injection (**Fig 3a**). As previously demonstrated, adoptively transferred DCs were detectable in the popliteal lymph node 18-24 h post-transfer and they largely disappeared by 48 h [17-19]. FlAsH signal remained stable on DCs for the entire duration DCs were present in the lymph node (**Fig 3b**). Strikingly, FlAsH was picked up by the cognate OTII T cells, suggestive of an antigen-specific peptide major histocompatibility complex II (pMHCII) trogocytosis in-vivo (**Fig 3c, d**). Antigen-specific transfer was also confirmed by OVA-Biotin staining on OTII T cells by fluorochrome conjugated streptavidin (**Fig 3c, d**). We had previously shown that antigen-specific Treg cells trogocytose pMHCII from DC synapse more efficiently than their effector CD4+ T cell counterparts [20]. To quantify FlAsH signal picked up by OVA$_{323\text{-}339}$-specific Treg cells, naïve OTII T cells were differentiated into induced Treg (iTreg) cells. While iTreg cells cocultured with antigen-pulsed DCs performed greater MHCII trogocytosis than naïve T cells at 18 h in vitro, peptide-specific trogocytosis was indiscernible within this time frame (**Supplementary figure 1a-b**). Similar to naïve OTII T cells, OTII iTreg cells acquired OVA$_{CACA}$ and OVA-Biotin, with distinct peptide-specific signal and proliferative response on day 2 post-adoptive transfer (**Supplementary figure 1c-f**).

Intravital two-photon microscopy of the popliteal lymph node further confirmed the antigenicity of OVA$_{CACA}$ in vivo by revealing the morphological changes compatible with T cell priming such as increased T cell volume and decreased sphericity (**Fig 3e**). OTII-DC synapses had greater 3D

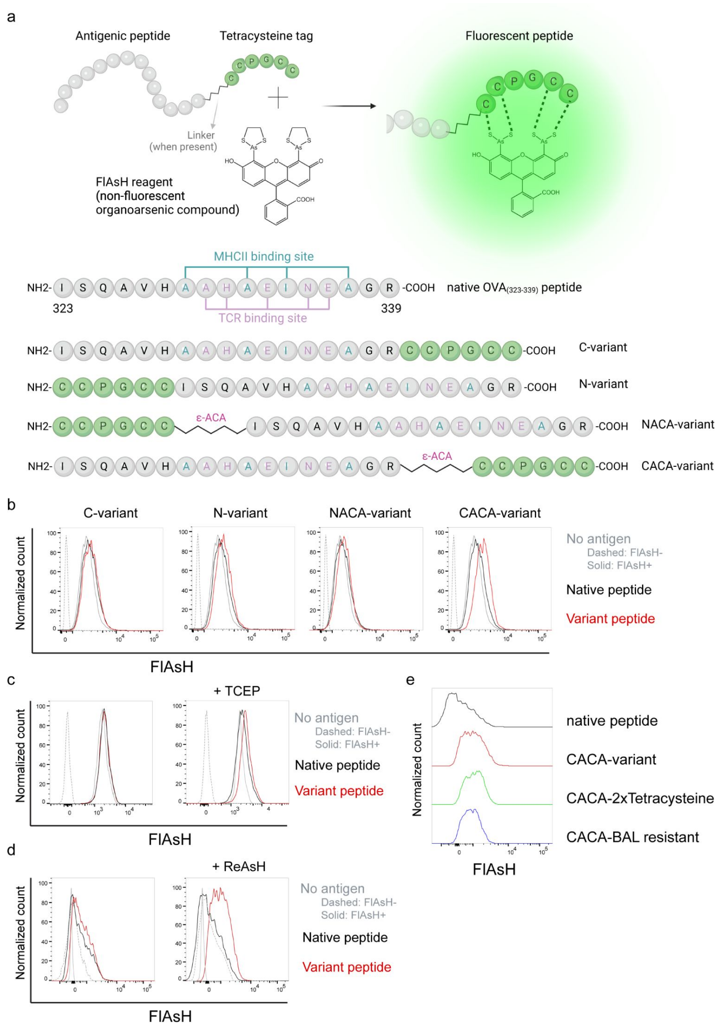


Figure 1

FlAsH labeling elicits fluorescence depending on the location of the tetracysteine tag.

a, AsH- probe labeling strategy for the tetracysteine-tagged peptides and the four $OVA_{323\text{-}339}$ variants generated for the study. **b**, FlAsH signal intensity on splenic DCs pulsed with tagged $OVA_{323\text{-}339}$ variants. **c**, FlAsH signal intensity on splenic DCs pulsed with CACA-variant following treatment with TCEP, a strong reducing agent. **d**, Signal intensity following pre-treatment of splenic DCs with ReAsH, followed by pulsing with CACA-variant and FlAsH treatment. **e**, FlAsH signal intensity of modified CACA-variant peptide with a duplicated CCPGCC sequence (CACA-2xTetracysteine) and CACA-variant peptide flanked with BAL-resistant peptide sequences (CACA-BAL resistant). Data in **b-d** are representative of three independent experiments, each performed with three biological replicates.

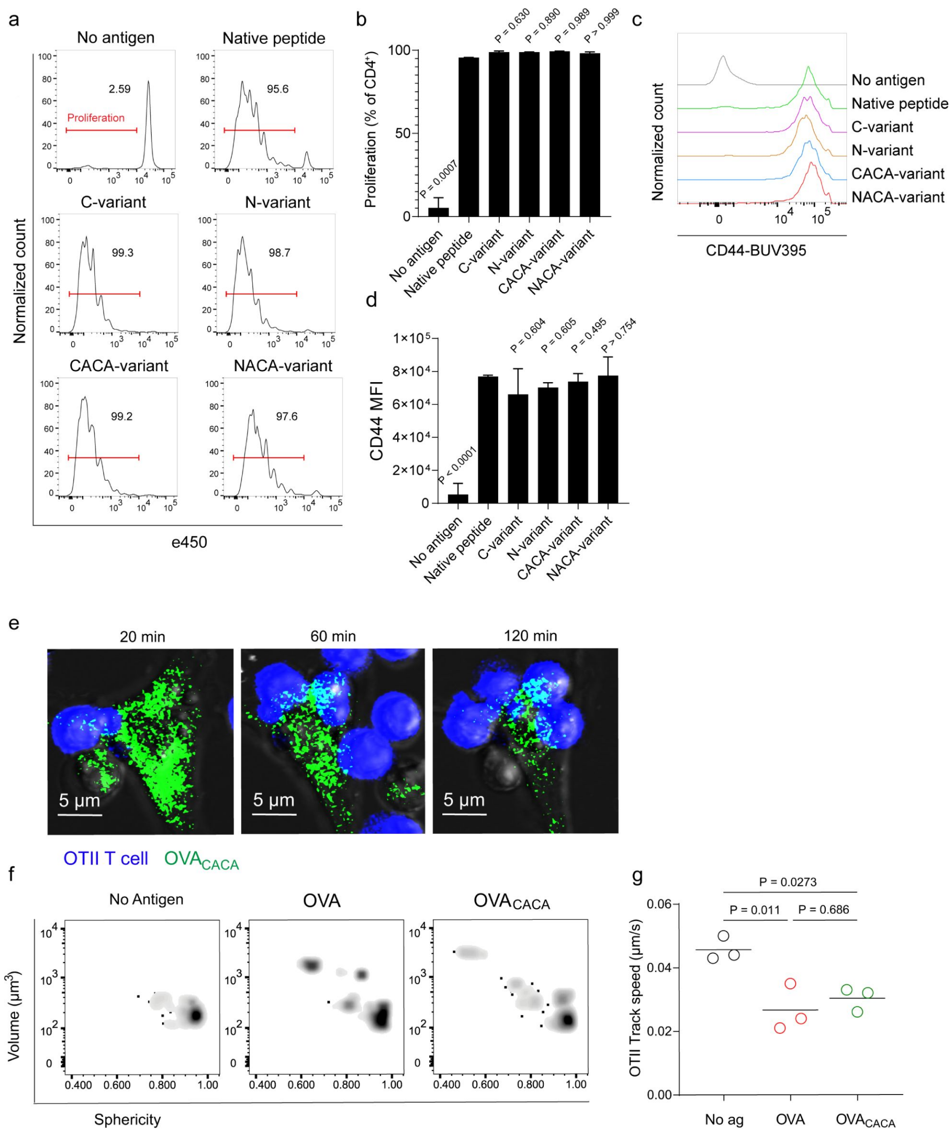


Figure 2

Tetracysteine tag does not alter the T cell priming ability of the native peptide.

a-d, DCs were loaded with the indicated peptides (5 μM) and co-cultured with e450-labeled naïve OT-II T cells (5×10^4) at a 1:10 DC to T cell ratio. **a**, Histograms (**a**) and the bar graph (**b**) demonstrate the proliferation status of T cells at 72 hours. Histograms (**c**) and the bar graph (**d**) depict the surface expression of CD44, a T cell activation marker. The bars (**b,d**) indicate the mean of three biological replicates; the error bars show the standard deviation. Data are representative of three independent experiments. P values refer to comparisons between each group and the native peptide, calculated using one-way ANOVA with Dunnett's multiple comparisons test. **e-g**, Peptide-pulsed DCs (2×10^5) were treated with FlAsH, cocultured 1:1 with e450-labeled naïve OT-II T cells and imaged for real-time interactions for 3 h (blue, OTII; green, FlAsH). **e**, Time lapse images depict the clustering of OTII T cells around OVA_{CACA}-pulsed DCs. **f**, Histocytometry plots for OTII T cell volume and sphericity. **g**, Graph shows the average track speed of OTII T cells. The lines mark the means of n = 3 replicates (one per symbol). Data are representative of two independent experiments. P values were calculated using a one-way ANOVA with Tukey's test.

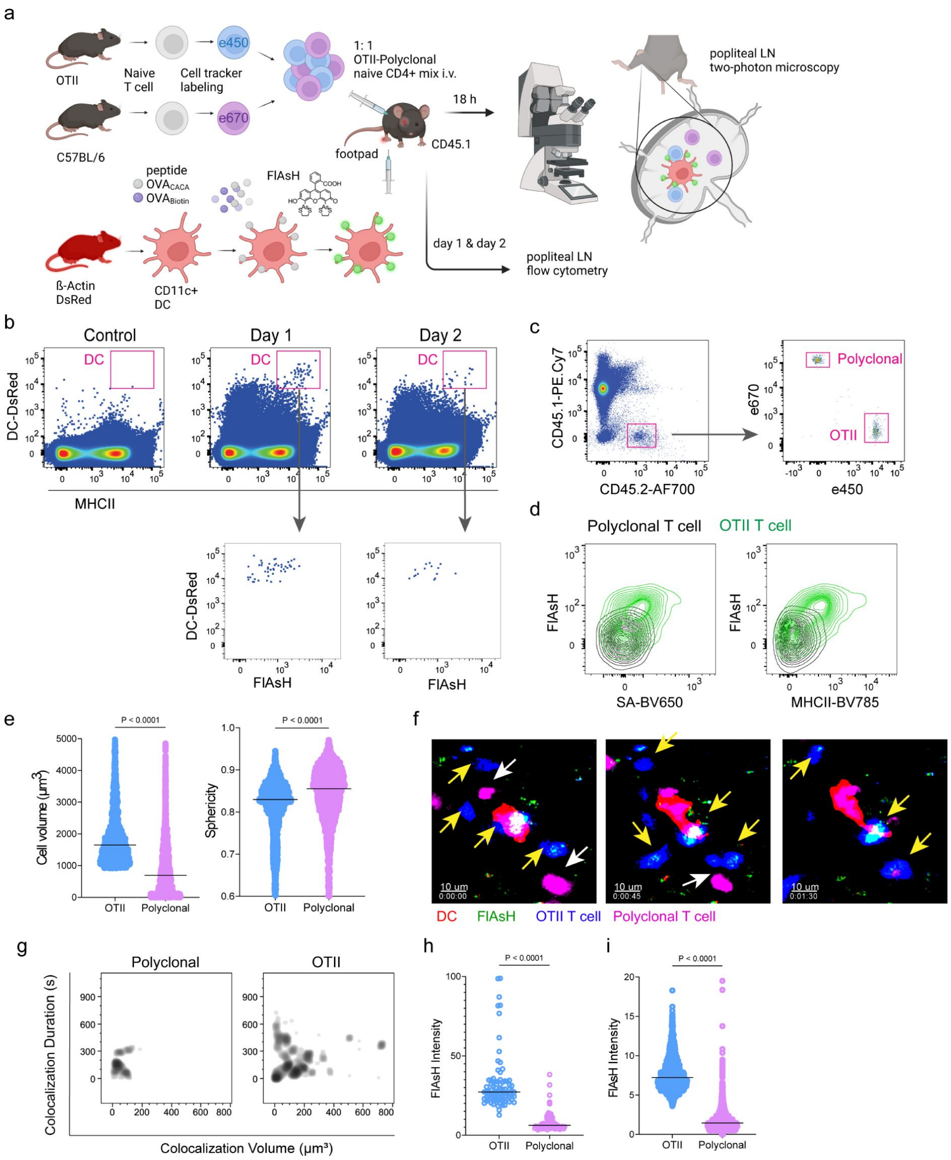


Figure 3

T cells acquire tetracysteine-tagged peptide from immune synapse.

DsRed+ splenic DCs were double-pulsed with 5 μM OVA_{CACA} and 5 μM OVA-biotin and adoptively transferred into CD45.1 recipients (2 × 10^6 cells) via footpad. Naïve e450-labeled OTII (1 × 10^6 cells) and e670-labeled polyclonal T cells (1 × 10^6 cells) were injected i.v. Popliteal lymph nodes were either imaged at 18 h or analyzed by flow cytometry at 18 h or 48 h post-transfer. **a**, Experimental approach. **b**, Flow cytometry plots gating on DsRed+ DCs over 48h (top panel) and demonstrating the OVA_{CACA} levels (FlAsH, bottom panel). **c**, Gating strategy for T cells of the recipient (CD45.1) and donor (CD45.2). **d**, T cell levels of OVA_{CACA} (FlAsH), MHCII, and OVA-biotin (detected with streptavidin) at 48 h following adoptive transfer. **a-d**, Data are representative of three independent experiments with n=3 mice per time point. **e-i**, Summary data from live microscopy showing OTII cell blasting (**e**), DC-T cell interaction with 45 min intervals (**f**), plots for the DC-T cell contact duration and volume, graphs for the FlAsH intensity limited to the DC-T cell contact regions (**g-h**) and average Flash intensity acquired by T cells (**i**). Data are representative of four independent experiments with n=2 mice per experiment. P values were calculated using two-sided Welch's t-test.

volume and duration and harbored more FlAsH than surrounding regions (**Fig 3g-h**). Overall, FlAsH signal was exclusively picked up by OTII T cells imaged, as early as 18 h (**Fig 3i**, *Supplementary video 1*).

AsH-probe robustly tracks pMHCII complexes

To delineate whether FlAsH labeling reports free vs. MHCII-bound peptide in cells, we initially assessed how surface MHCII expression of cells correlates with the OVA_{CACA}-FlAsH signal. C57BL/6 splenocytes were incubated with OVA-Biotin or OVA_{CACA} and peptide distributions among the T cells, B cells and DCs were quantified by flow cytometry using streptavidin or FlAsH (**Fig 4a**). Both streptavidin and FlAsH signals correlated significantly with MHCII expression, providing the strongest signal in DCs, where $CD4^+$ T cells served as a negative control due to lack of MHCII expression in mouse T cells [21] (**Fig. 4b**). It is worth mentioning that streptavidin detection of biotinylated peptide is a viable option for quantifying peptide loading on cells. However, size and membrane impermeable nature of streptavidin make it unsuitable for assays that include live monitoring of the T-DC interaction [22]. Next, we addressed whether FlAsH labeling of OVA_{CACA}-pulsed DCs is dependent on the MHC expression. $CD11c^+$ DCs were isolated from MHCII single knockout or MHCI and II double knockout (KO) mice, labeled with tracking dyes and mixed 1:1 with WT C57BL/6 DCs. Subsequently, they were incubated with OVA_{CACA} and treated with ReAsH and FlAsH as in **Fig.1** (**Fig 4e**). To make a fair comparison between WT and KO DCs and minimize autofluorescence, we performed adoptive transfers, aiming to facilitate in vivo elimination of apoptotic and possibly defective KO DCs [23, 24]. Flow cytometry analysis of the draining lymph node demonstrated that WT DCs, but not MHCII KO DCs, harbored FlAsH. Two-photon microscopy of the lymph node further confirmed these findings, indicating that FlAsH robustly marks OVA_{CACA}-MHCII complexes (**Fig 4e-h**).

To address how FlAsH acquisition from synapse marks antigen-specific vs. bystander T cells, DCs were labeled with ReAsH and cell membrane dye PKH-26, pulsed with OVA_{CACA}, $E\alpha_{(52\text{-}68)}$ or left unpulsed. Next, they were treated with FlAsH and injected into the recipient mice via footpad. Subsequently, 1:1 mixture of labeled naïve OTII and Eα-specific T cells were adoptively transferred into via retroorbital sinus i.v. and the popliteal lymph nodes of the recipients were removed 36-42 h after the transfer (**Fig. 5a**). Flow cytometry analysis showed that OTII and Eα-specific T cells both acquired PKH-26 from double-pulsed DCs, indicating that they captured DC cell membrane from synapse (**Fig 5b**). This process is termed as TCR-mediated trogocytosis and has previously been described as a feature of the DC-T cell synapse [11]. Although both T cell types trogocytosed double-pulsed DC cell membrane, FlAsH signal was picked up only by OTII T cells, suggesting that FlAsH tracks pMHCII mobility with a high precision (**Fig 5c**).

Furthermore, we visualized dynamic $CD4^+$ T cell-DC contact and pMHCII mobility in vivo at 18 h after the transfer. DCs were labeled with ReAsH and pulsed with OVA_{CACA} and/or LCMV $GP_{(61\text{-}80)}$ and treated with FlAsH. Recipient mice were injected with unpulsed, OVA_{CACA} single or OVA_{CACA}-$GP_{(61\text{-}80)}$ double pulsed DCs via footpad along with a 1:1 mixture of labeled naïve OTII and SMARTA T cells i.v. via retroorbital sinus. Popliteal lymph node microscopy confirmed that OTII and SMARTA T cells had stable contacts with double-pulsed DCs (**Fig 5d**, *Supplementary video 2*). Similar to Eα-specific T cells, SMARTA also failed to pick up FlAsH signal despite showing signs of antigen-specific contact such as increased volume, reduced sphericity and decreased track speed (**Fig 5 e-f**).

Tetracysteine-tagging of peptides provides a sensitive tool to visualize interactions with altered TCR affinity

The lack of adaptable antigen-targeted probes presents a significant obstacle in the dynamic visualization of antigen-specific immunity. To address this gap, we sought to determine robustness of tetracysteine tagging in visualizing nuanced pMHCII variants. We created three variants by introducing alanine mutations at the TCR binding residues 331, 333, and 335 of OVA_{CACA} (**Fig**

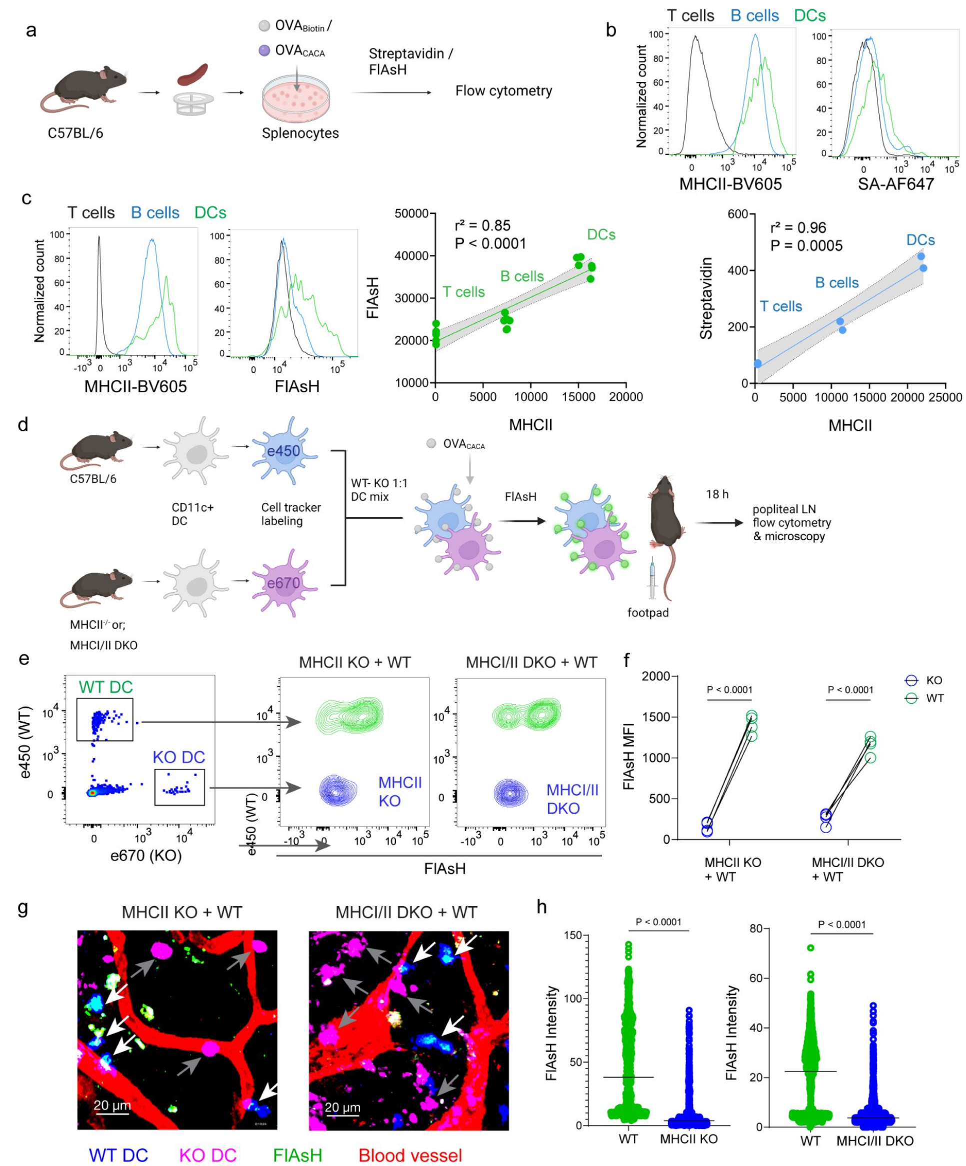


Figure 4

AsH probe reports the peptide loaded on the MHCII.

a-c, Relationship between surface MHCII expression and peptide load. **a**, Experimental scheme. **b**, Plots (top) show the surface levels of MHCII and OVA-biotin, graph (bottom) shows their correlation. Data are representative of two independent experiments performed with n = 2 biological replicates (one per symbol). **c**, Surface levels of MHCII and OVA_{CACA} (FlAsH) and their correlation. Data are representative of two independent experiments performed with n = 6 biological replicates (one per symbol). Simple linear regression and Pearson correlation and were used to demonstrate the relationship between surface MHCII and peptide load, gray area and dotted lines mark the standard error of the fitted line (**b,c**). **d**, Schematic depicting in vivo adoptive transfer approach. WT and KO DCs were labeled, mixed, pulsed with 5 μM OVA_{CACA}, labeled with FlAsH and adoptively transferred to WT recipients via footpad (1-2 × 10^6 cells). **e-f**, Flow cytometry plots (**e**) and graphs (**f**) showing FlAsH intensity of WT, MHCII-KO, or MHCI/II double-KO DCs that had migrated to the lymph node within 18 h following adoptive transfer. Data are representative of two independent experiments with n=4 mice per group. P values were calculated using two-sided Student's t-test. **g-h**, Microscopy images of popliteal lymph node following anti-CD31 (red) antibody injection i.v., yellow arrows point to WT, white arrows point to KO DCs (**g**), FlAsH intensity of adoptively transferred DCs (**h**). Data are representative of four independent experiments with n=2 mice per experiment. P values were calculated using two-sided Welch's t-test.

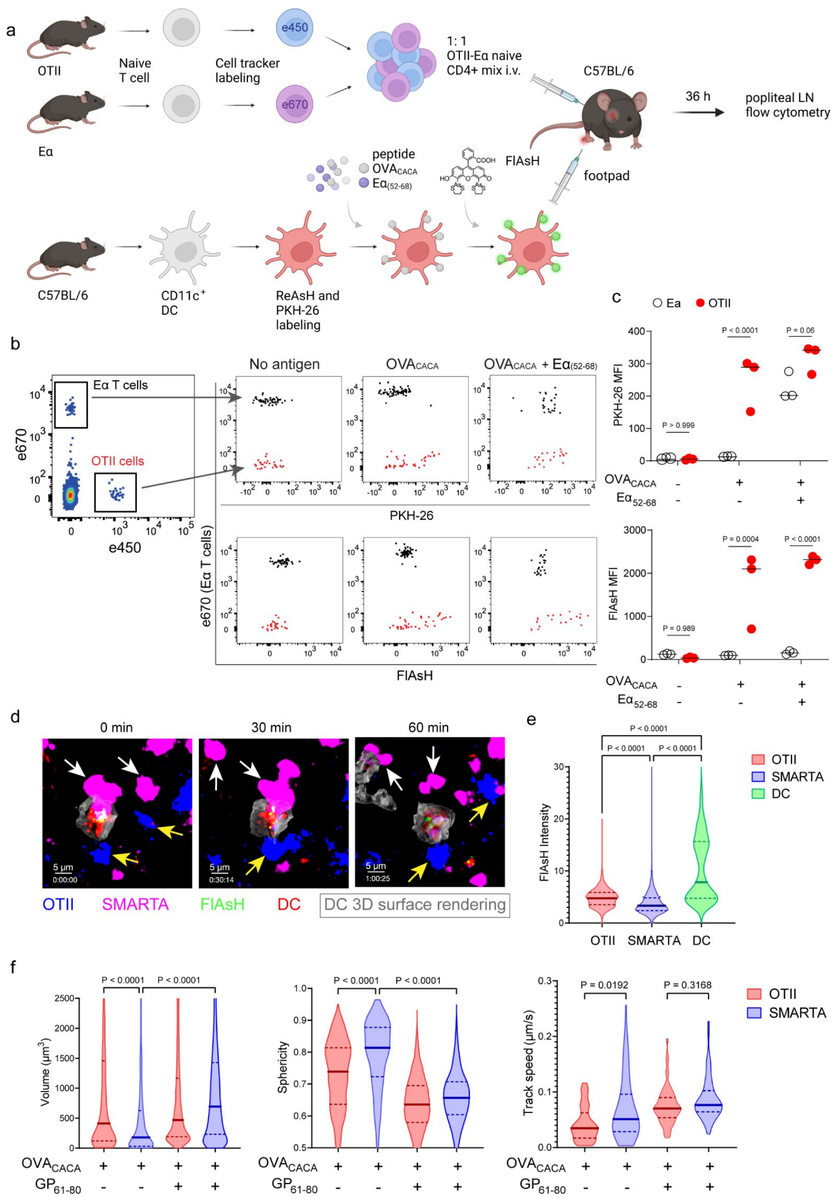


Figure 5

FlAsH-pMHCII is incorporated by cognate T cells in an antigen-specific manner.

a-c, DCs were labeled with ReAsH and PKH-26, single/ double-pulsed with 5 μM OVA_{CACA}, $E\alpha_{(52\text{-}68)}$ or left unpulsed. They were labeled with FlAsH and adoptively transferred into WT mice (0.5-1 × 10^6 cells) via footpad. Naïve e450-labeled OTII and e670-labeled Eα-specific T cells were mixed 1:1 (0.25-0.5 × 10^6/ T cell type) and injected i.v. Popliteal lymph node was analyzed by flow cytometry at 36-42 h post-transfer. **a**, Experimental scheme. **b**, Summary plots. **c**, PKH-26 and FlAsH MFI in OTII and Eα-specific T cells. Data are representative of two independent experiments with n=3 mice per group. P values were calculated using two-way ANOVA with Sidak's multiple comparisons test. **d-f**, DCs were labeled with ReAsH and single/ double-pulsed with 5 μM OVA_{CACA}, LCMV $GP_{61\text{-}80}$, labeled with FlAsH and adoptively transferred into WT mice (1-2 × 10^6 cells) via footpad. Naïve e450-labeled OTII and e670-labeled SMARTA T cells were mixed 1:1 (0.5-1 × 10^6/ T cell type) and injected i.v. Live popliteal lymph node sections were imaged at 18 h. **d**, Time series representative of DC-T cell interactions. **e**, Average Flash intensity of the adoptively transferred cells. **f**, T cell blasting (left and middle panels) and motility (right) following adoptive transfer with DCs pulsed with the indicated peptides. Data are representative of three independent experiments with n=2 mice per group. P values were calculated using one-way ANOVA with Tukey's multiple comparisons test.

6a). The mutations at 331 and 333, depicted as variants 1 and 2 respectively, were previously reported to abolish the binding of OTII TCR to $OVA_{323-339}$ [9]. Although the residue 335 is flanked between the core OTII TCR binding residues 333 and 336, mutant 335 was shown to elicit normal proliferative response in vitro [9]. We addressed how pMHCII uptake by T cells is influenced by the mutation of TCR binding residues of OVA_{CACA}. DCs were isolated from CD45.1 congenic mice, loaded with variant peptides and cocultured with naïve CD45.2 OTII T cells for three days (**Fig 6b-e**). In line with earlier observations, variants 1 and 2 interrupted T cell priming while variant 3 led to a proliferative response similar to native peptide (**Fig 6c-d**). TCR mediated uptake of the FlAsH-labeled pMHCII complexes followed the same pattern and only variant 3 complexes produced a comparable signal to that of OVA_{CACA} (**Fig 6e**). These suggest that AsH-probe labeling can be a sensitive method to track low affinity/ nuanced pMHCII complexes.

Discussion

Dynamic monitoring of antigen-specific T cell responses in vivo poses challenges due to the limited availability of reagents and the size constraints of the immune synapse. Here, we address this issue by presenting a novel strategy for visualizing T cell-APC interactions via tetracysteine tagging of antigenic peptides. While tetracysteine tagging has been previously applied in cell biology and virology, these applications have been mostly limited to cell lines and in vitro studies in immunology unrelated experimental settings [25-28]. On the other hand, our technique labels antigenic peptides and tracks pMHCII complexes in primary APCs during their interactions with T cells both in vitro and in vivo. Our approach is superior as compared to other strategies for monitoring T cell-DC interactions, in the way it robustly targets specific pMHCII complexes in real time in vivo. There is no need for special organisms or genetic modifications, and the technique can easily be expanded to new antigens and T cell-APC pairs from both mice and humans. This strategy can, for instance, be integrated into validation efforts for antigen discovery and utilized in the investigation of the synapses of newly deorphanized human T cells, a largely unexplored area where tools are scarce [29-32]. Given that is a two-part system (tag and FlAsH), it also carries the added capability of being inducible at different time points to allow for increased flexibility in interrogating pMHCII-TCR interactions over time. On the other hand, this introduces a high background due to the non-specific binding of FlAsH to endogenous cysteines. Thus, we implemented a ReAsH pretreatment strategy, successfully improving the signal-to-noise profile for FlAsH. It's worth mentioning our unpublished attempts to load dendritic cells (DCs) with preconjugated peptide and FlAsH to mitigate the background. Unfortunately, this endeavor led to almost a complete loss of the FlAsH signal and has not been pursued further. Therefore, exploring new peptide probes chemically conjugated to small organic dyes emerges as a potential avenue for future investigation.

Most important of all, our experiments demonstrate that tetracysteine-tagged peptides can be transferred from DCs to T cells via synaptic processes, such as antigen-specific pMHCII trogocytosis/transendocytosis, marking acceptor T cells for several days. Although the turnover of pMHCII in acceptor T cells and its temporal relationship to the TCR downstream events require further elucidation, our visualization of tetracysteine-tagged pMHCII can complement tetramers in detecting antigen-specific T cells in vivo. Additionally, we show that tetracysteine tagging of pMHCII does not broadly alter or disturb T cell-DC synapse by functional read-outs. Although the ratio of peptide-FlAsH conjugates to free peptide in DCs would be needed to support this claim, our approach offers an advantage over tetramers in that it is suitable for real-time monitoring of the immune synapse. Finally, it is worth noting that tetracysteine tags can be visualized as electron-dense particles by Transmission Electron Microscopy [33, 34], providing potential for the detailed investigation of T-DC synapse and subcellular localization of pMHCII complexes. Although TEM application was out of scope for this study, it offers a promising avenue for future research.

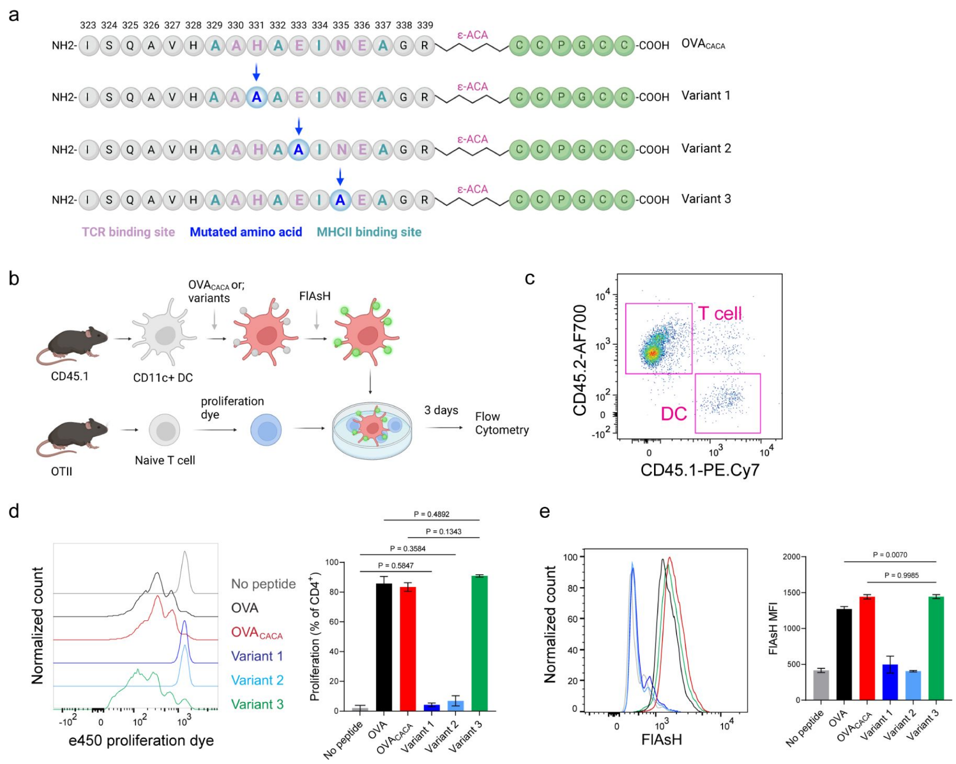


Figure 6

Nuanced TCR-pMHCII interactions can be discerned by FlAsH labeling of peptides.

a-b, Graphical depiction of the alanine-mutants for OVA_{CACA} TCR binding site and experimental approach. 0,5 × 10^5 $CD45.1^+$ DCs were pulsed with the indicated peptides, labeled with FlAsH and cocultured with 2,5 × 10^5 e450-labeled OTII cells for 3 days. **c**, Gating strategy for T cells and DCs. **d**, Representative flow cytometry plots for OTII T cell proliferation with a partial agonist (top) and two antagonist peptides (bottom). **e**, FlAsH signal intensity of DCs and OTII cells with partial agonist (top) and antagonist peptides (bottom) adoptive transfer with the indicated peptides followed by FlAsH treatment. Data are representative of two independent experiments performed with n = 3 biological replicates.

In conclusion, the tetracysteine labeling approach described here provides a powerful and unique tool to advance our understanding of T cell-APC interactions, offering new opportunities to investigate the immune response and develop improved immunotherapies.

Materials and Methods

Animals and reagents

C57BL/6, DsRed.T3, *H2-K1/H2-D1* knockout (MHCI$^{-/-}$), E$\alpha_{52\text{-}68}$, LCMV $GP_{61–80}$-specific (SMARTA) and $OVA_{323–339}$-specific (OT-II) TCR transgenic mice were acquired from The Jackson Laboratory. H2-Ab1 knockout (MHCII$^{-/-}$) mice were obtained from Taconic Biosciences. *H2-K1/D1*$^{-/-}$H2-Ab1$^{-/-}$ (MHCI-II double knockout) mice were obtained by breeding single knockout mice to homozygosity. All mice were maintained in Ohio State University ULAR and National Institutes of Health (NIH) animal facilities in compliance with Institutional Animal Care and Use Committee standards.

Cell cultures were maintained in sterile complete Roswell Park Memorial Institute (RPMI) media. The RPMI 1640 medium was supplemented with 10% heat-inactivated fetal bovine serum, 50 U/ml penicillin, 50 µM streptomycin, 1 mM sodium pyruvate, 2 mM L-glutamine, 0.1 mM non-essential amino acids, 50 µM 2-mercaptoethanol, and 10 mM HEPES (Thermo Fisher Scientific). For magnetic separations, cells were maintained in filtered and degassed magnetic-activated cell sorting (MACS) buffer. The MACS buffer consisted of PBS (Lonza) supplemented with 0.5% BSA (Sigma-Aldrich) and 2 mM EDTA (Sigma-Aldrich). For splenic dendritic cell (DC) isolation, Liberase TM (Blendzyme) and DNase I were purchased from Roche. 50 mg Liberase was dissolved in complete RPMI, resulting in a final concentration of 5 mg/mL. The solution was kept on ice, stirred gently every 5 minutes for 30 minutes and aliquoted into 1 mL portions. 10 mg/mL DNAse stock solution was prepared with 0.15 M NaCl and divided into 500 µL aliquots. Both enzymes were stored at -20ºC.

Samples were stained for flow cytometry using fluorescence-activated cell sorting (FACS) buffer (either PBS or Hanks' balanced salt solution (HBSS) supplemented with 2% fetal bovine serum, 1% HEPES, and 10 mM sodium azide (Sigma-Aldrich)). For confocal microscopy, PBS supplemented with 1% BSA was used. The antibodies used for flow cytometry are as follows: anti-CD4-efluor450 (clone GK1.5), anti-B220-PE (clone RA3-6B2), anti-CD11c-AF700 (clone N418), anti-CD45.1-PE.Cy7 (clone A20, Thermo Fisher Scientific), anti-CD45.2-AF700 (clone 104), anti-CD44-BUV395 (clone IM7, BD Biosciences), anti-I-A/I-E-BV605 (clone M5/114.15.2), anti-I-A/I-E-BV785 (clone M5/114.15.2), streptavidin-BV650, streptavidin-AF647 and purified anti-CD16/32 (clone 93). The antibodies were purchased from BioLegend unless stated otherwise. All peptides used in the study (*Supplementary Table 1*) were obtained from the NIH Research Technologies Branch, NIAID Peptide Core Facility.

For AsH probe labeling, peptides were pre-treated with fresh Tris Carboxy Ethyl Phosphene (TCEP, Sigma-Aldrich) to protect tetracysteine residues from oxidation. 10 mM ReAsH stock solution was prepared by adding 45 µL sterile cell-culture grade DMSO (Sigma-Aldrich) to 250 µg solid ReAsH (Cayman Chemical) and 10 mM FlAsH solution was prepared by adding 150uL DMSO to 1 mg FlAsH powder (Cayman Chemical). British Anti-Lewisite (BAL, Alfa Aesar) and HBSS with Ca^{++} and Mg^{++} (Thermo Fisher Scientific) was used to wash excess FlAsH off the cells. BAL is resuspended in sterile degassed water to obtain 8 mM stock solution. FlAsH, ReAsH and BAL are highly sensitive to oxidation, therefore their activity may decline in subsequent uses. This may be averted, at least in part, by quickly purging stock solution containers with 4M Argon gas (Sigma-Aldrich, 501247), placing container in a 50mL tube, purging again with Argon before storing FlAsH and ReAsH at -20ºC, BAL at 4ºC. We do not recommend storage longer than one month or repeated use of the same reagent stock.

DC isolation, peptide loading and -AsH probe labeling

Mature splenic DCs were isolated as previously described [35] and resuspended in complete RPMI. Peptides were added into 20 mM TCEP solution at twice the desired concentration for loading DCs. The solution was vortexed thoroughly and incubated at 37ºC for 30 minutes. Meanwhile, ReAsH stock solution was diluted to 100 uM in complete RPMI and added on DC suspension to obtain 1 uM final ReAsH concentration. Cells were incubated at 37ºC for 30 minutes and washed twice with complete RPMI. ReAsH labeled DCs were mixed with equal volume of peptide-TCEP solution, incubated at 37ºC for 30 minutes, washed twice and resuspended in complete RPMI to proceed with FlAsH labeling. Flash was added to cell suspension to achieve 1 uM final concentration, incubated at 37ºC for 30 minutes and washed twice with complete RPMI. To remove excess FlAsH and reduce non-specific binding, cells were resuspended in 240 uM BAL diluted in HBSS with Ca^{++} and Mg^{++}, washed twice, then washed a third time with PBS and resuspended in PBS.

DC-T cell cocultures

Freshly isolated DCs were pulsed with peptide in complete RPMI and were incubated at 37 °C for 30 min followed by three washes with complete RPMI to remove unbound peptide. Pulsed DCs were then treated with -AsH as decribed above then seeded into flat-bottom 96-well plates (Corning) at a density of 5 × 104 to 1 × 105 cells per well. T cells and iTreg cells were labeled with e450 or e670 (eBioscience) according to the manufacturer's protocol. Cells were cocultured at 37°C for for indicated periods, at a 1:10 ratio of DCs to T cells for proliferation assays and 1:1 for trogocytosis assays.

iTreg differentiation

For iTreg differentiation, Tnaive cells were isolated and 24-well sterile tissue culture plates (Corning) were coated with anti-CD3ε (BioLegend) and anti-CD28 (BioLegend) as described [36]. Naïve $CD4^+$ T cells were resuspended in complete RPMI media supplemented with 100 IU ml−1 recombinant human IL-2 (Peprotech) and 5 ng ml−1 recombinant human TGF-β (Peprotech). 3 × 10^5 cells were added to the wells at a volume of 1 ml per well. Cells were cultured at 37 °C, 5% CO2 for 3-4 days.

Trogocytosis assay

Trogocytosis assays were perfomed as described in [20]. Briefly, freshly isolated splenic DCs were labeled with 4μM PKH26 (Sigma-Aldrich) for labeling of the cell membrane as described in Puaux et al.48. DCs were then pulsed with 10 μM peptide, treated with -AsH and were cocultured with T cells for up to 18 h. Cell conjugates were dissociated by washing cells with MACS buffer with 2 mM EDTA; cell suspensions were prepared for flow cytometry.

Flow cytometry

Cells were stained in FACS buffer containing fluorochrome-conjugated antibodies for 30 minutes at 4°C, protected from light. Data acquisition was performed with BD LSRFortessa and BD LSR II cytometers (BD Biosciences). Data analysis was performed with FlowJo v10.8.1.

Adoptive transfer

Mice were anesthetized in an induction chamber with 1.5% isoflurane USP (Baxter). Pulsed DCs were adoptively transferred through the right footpad in 50uL sterile PBS. T cells labelled with e450 or e670 (Thermo Fisher Scientific) were injected intravenously in 100uL sterile PBS through the retro-orbital sinus.

Confocal microscopy

35mm dishes with a 14mm glass coverslip bottom (MatTek) were pre-treated with 10ug/ml fibronectin (Sigma-Aldrich) in PBS for 1h at room temperature and washed twice. Splenic DCs were added onto the glass coverslip in complete RPMI and pulsed with 5-10 μM peptide at 37°C for 1 hour. Excess peptide was washed off with complete RPMI before addition of CD4 T cells. Co-cultures were imaged for 3-18h using a Leica SP-8 inverted microscope (Leica Microsystems) equipped with a full range of visible lasers, two hybrid detectors, three photomultiplier detectors, and a motorized stage. 63× objective (Leica Microsystems) was used and microscope configuration was set up for 3D analysis (x, y, z) of the cellular layer. The following lasers were used: diode laser for 405 nm excitation; argon laser for 488 and 514 nm excitation; diode-pumped solid-state laser for 561 nm; and HeNe laser for 594 and 633 nm excitation. All lasers were tuned to minimal power (between 0.3% and 2%) to prevent photobleaching. z stacking of images of 10–12 μm were collected. Mosaic images of large cell culture areas (1 mm2) were generated by acquiring multiple z stacks using the Tile scan mode and were assembled into tiled images using LAS X v4.0 (Leica Microsystems).

Intravital two-photon laser-scanning microscopy of mouse popliteal lymph nodes

Mice were anesthetized in 1.5% isoflurane USP (Baxter) through a rodent nose cone followed by surgery to expose the right popliteal lymph node. Warm PBS was added to maintain lymph node moisture. Mouse body temperature was maintained with an infrared blanket (Braintree Scientific) throughout the imaging process after which they euthanized through cervical dislocation under anesthesia. Mice were imaged on a glass stage with a two-photon laser-scanning microscopy setup with a Leica SP8 inverted confocal microscope with dual multi-photon lasers, Mai Tai and InSight DS (Spectra-Physics), L 25.0 water-immersion objective, 0.95 NA (Leica Microsystems), and a 37 °C incubation chamber (NIH, Division of Scientific Equipment and Instrumentation Services). The Mai Tai laser was tuned to 890 nm to excite e450and FlAsH; the InSight DS laser was tuned to 1,150 nm to excite DsRed/ PKH-26/ ReAsH and e670. For time-lapse imaging, a z stack consisting of 10–12 single planes (5 μm each over a total tissue depth of 50–60 μm) was acquired at 15 second intervals for a total period of 1-4 h.

Two-photon laser-scanning microscopy of live lymph node sections

Live lymph node sections were imaged ex vivo by two-photon microscopy. To maintain tissue architecture, popliteal lymph nodes were immediately transferred to PBS with 1% BSA and kept on ice immediately following harvest. Residual fat and connective tissue were removed using a Leica MZ6 StereoZoom microscope (Leica Microsystems). Lymph nodes were suspended in 38 °C agarose in DMEM and kept over ice. Following complete agarose solidification, complete RPMI was added and blocks were cut containing one lymph node each. Lymph node blocks were cut into 200 μm sections with a Leica VT1000 S vibrating blade microtome (Leica Microsystems) at speed 5 in ice-cold PBS and frozen at -80 °C. Prior to imaging, complete RPMI medium was added to tissue sections and incubated at 37 °C for 2 h. Sections were held down with tissue anchors (Warner Instruments) in 35mm dishes with a 14mm glass coverslip bottom (MatTek) and were imaged with a Leica SP8 inverted five-channel confocal microscope equipped with an environmental chamber (NIH, Division of Scientific Equipment and Instrumentation Services) and a motorized stage. The microscope configuration was set up for four-dimensional analysis (x, y, z, t) of cell segregation and migration within tissue sections. A diode laser for 405 nm, an argon laser for 488 and 514 nm, a diode-pumped solid-state laser for 561 nm, and a HeNe laser for 594 and 633 nm excitation wavelengths were tuned to minimal power (between 0.3% and 2%). z-stack of images of 10–25 μm were collected. Mosaic images of whole lymph nodes were generated by acquiring multiple z stacks using a motorized stage to cover the whole lymph node area and assembled into a tiled image with the LAS X software. For time-lapse analysis of cell migration, tiled z stacks were collected over time (1–4 h).

Image analysis

Analysis of confocal and intravital two-photon images was performed using Imaris v9.2.1 and v10.0.0 (Bitplane) as previously described [35]. In brief, e450 and e670 signals were utilized for reconstructing the 3D structure of T cells and iTreg cells as surface objects. To reconstruct the 3D structure of DCs, overlapping DsRed, PKH-26, and ReAsH signals were employed. Imaris v10.0.0's built-in algorithms were used to calculate 3D volumes, track velocities, surface-to-surface colocalization parameters, and FlAsH (green) intensities of cells.

Statistical analysis

Statistical analysis was conducted on Prism v9.5.1 (GraphPad). Statistical tests and relevant P values for each figure are indicated in the figure or figure legend.

Acknowledgements

This study was funded by was by the National Institutes of Allergy and Infectious Diseases Intramural Research Grant ZIAAI000224, Extramural Research Grant DP2AI154451 and research funds from the Ohio State University College of Medicine.

Declarations

Author Contributions

B.A. conceived the project; B.A. and M.A. designed the experiments and wrote the manuscript; M.A., J.A.S., D.W., R.K., and O.K. performed the experiments; B.A. and J.K. analyzed the data; E.M.S., J.A.S. and, D.W. edited the manuscript.

Supplementary Files

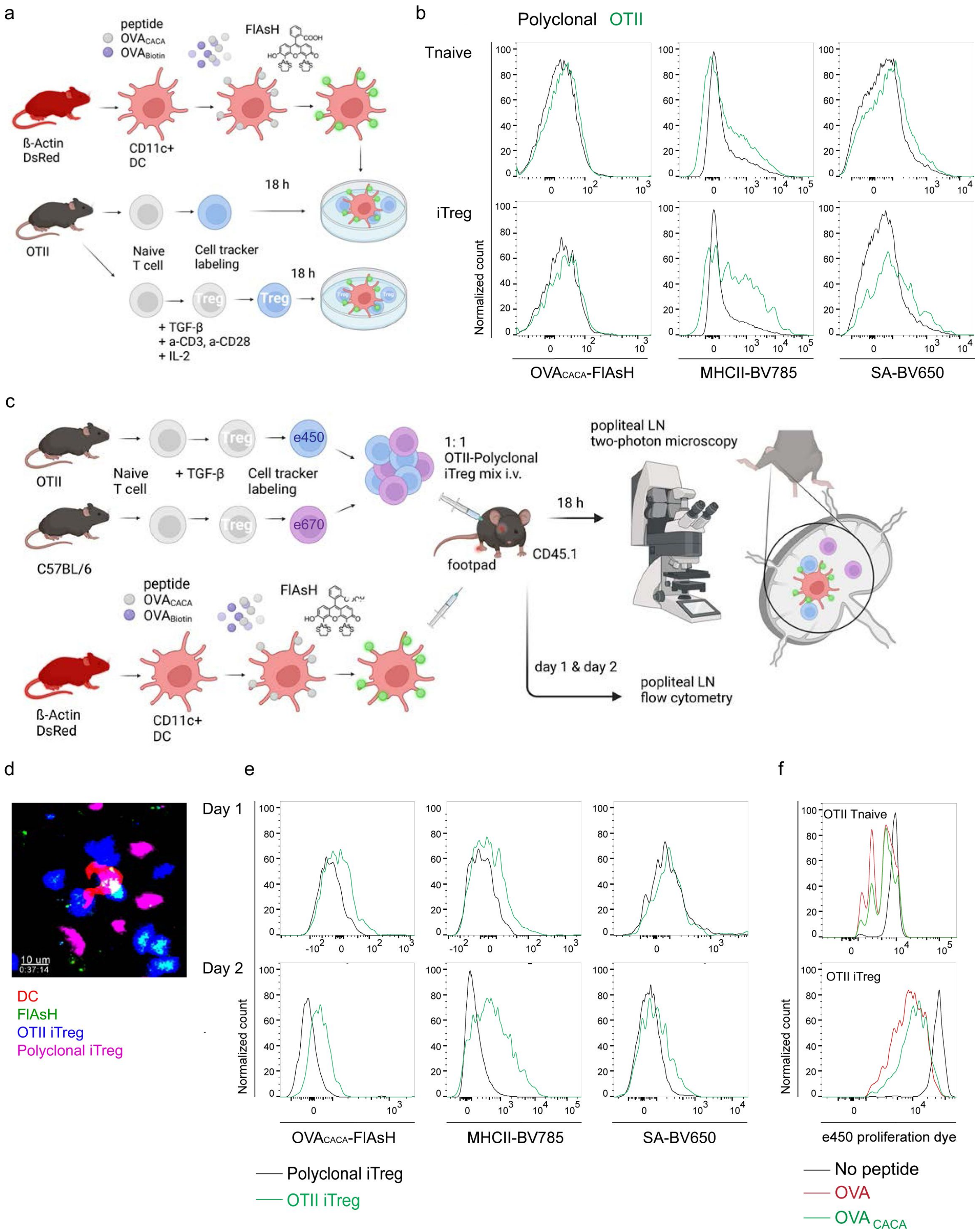
3193191v2_video1.mp4

3193191v2_video2.mp4

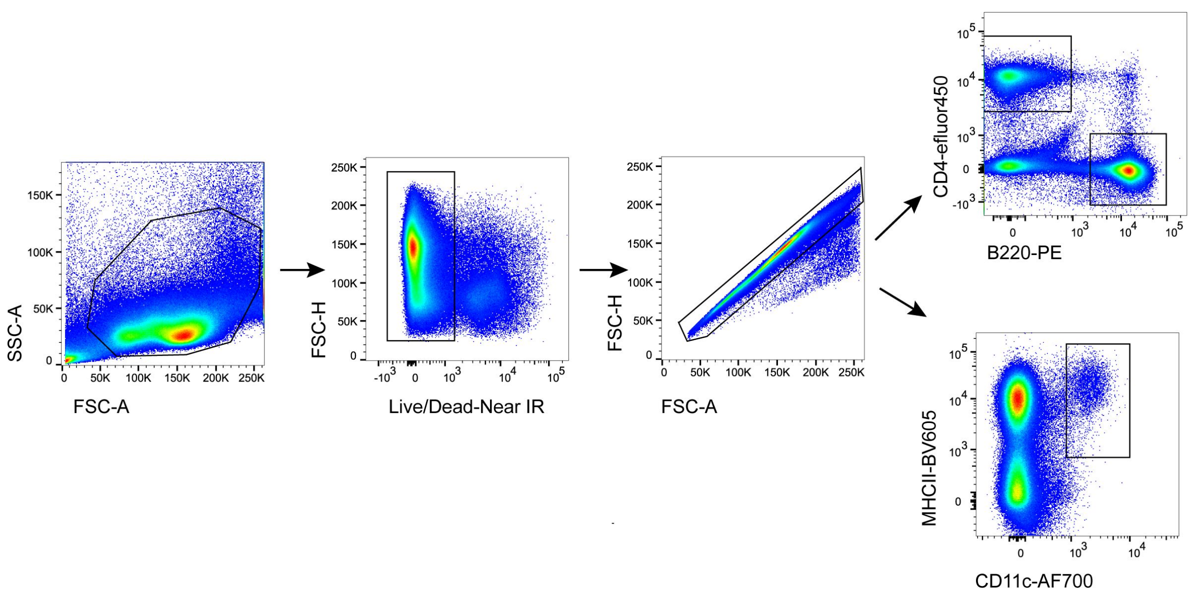
Supplementary table 1

reagent	MW	Sequence	Notes
OVA-4C-C	2340.65	ISQAVHAAHAEINEAGRCCPGCC	
OVA-4C-N	2340.65	CCPGCCISQAVHAAHAEINEAGR	
OVA-C-ACA	2453.81	ISQAVHAAHAEINEAGR-εACA-CCPGCC	
OVA-N-ACA	2453.81	CCPGCC-εACA-ISQAVHAAHAEINEAGR	
OVA-C-ACA x2	3020.53	ISQAVHAAHAEINEAGR-εACA-CCPGCCCCPGCC	x2
OVA-C-ACA (Re)	3185.95	ISQAVHAAHAEINEAGR-εACA-FLNCCPGCCMEP	BAL resistant
OVA-N-ACA (Re)	3185.95	FLNCCPGCCMEP-εACA-ISQAVHAAHAEINEAGR	BAL resistant
OVA-biotin	2000.4	ISQAVHAAHAEINEAGR-biotin	
OVA-C-ACA variant 1	2387.9	ISQAVHAAAAEINEAGR-εACA-CCPGCC	modified TCR binding site
OVA-C-ACA variant 2	2395.5	ISQAVHAAHAAINEAGR-εACA-CCPGCC	modified TCR binding site
OVA-C-ACA variant 3	2409.6	ISQAVHAAHAEIAEAGR-εACA-CCPGCC	modified TCR binding site

Supplementary Figure 1



Supplementary Figure 2



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Reviewer #1 (Public Review):

Summary:
The authors develop a method to fluorescently tagged peptides loaded onto dendritic cells using a three step method. These include a pre blocking step to block endogenous cysteine motifs on the DC surface, loading a tetracystein motif modified peptide on surface MHC and a labelling step done on the surface of live DC using a dye with high affinity for the added motif. The results are convincing in demonstrating in vitro and in vivo T cell activation and efficient label transfer to specific T cells in vivo. The label transfer technique will be useful to identify T cell that have recognised a DC presenting a specific peptide antigen to allow the isolation of the T cell and cloning of its TCR subunits, for example. It may also be useful as a general assay for in vitro or in vivo T-DC communication that can allow detection of genetic or chemical modulators.

Strengths:
The study include both in vitro and in vivo analysis including flow cytometry and two photon laser scanning microscopy. The results are convincing and the level of T cell labelling with the fluorescent pMHC is surprisingly robust and suggests that the approach is potentially revealing something about fundamental mechanisms beyond the state of the art. They also provide practical information about the challenges of the method and discuss limitations.

Weaknesses:
The method is demonstrated only at high pMHC density and it would need to be re-optimised to determine if it can be used at lower densities that may often be encountered physiologically.

- https://doi.org/10.7554/eLife.91809.2.sa1

Reviewer #2 (Public Review):

Summary:

The authors have developed novel Ovalbumin model (OTII) peptide that can be labeled with a site-specific FlAsH dye to track agonist peptides both in vitro and in vivo. The utility of this tool could allow better tracking of activated polyclonal T cells particularly in novel systems. The authors have provided solid evidence that peptides are functional, capable of activating OTII T cells, and these peptides can undergo trogocytosis by cognate T cells only.

Strengths:
-An extensive array of in vitro and in vivo studies are used to assess peptide functionality.
-Nice use of cutting edge intravital imaging,
-internal controls such as multiple non-cogate T cells were used to improve robustness of the results
-One of the strengths is the direct labeling of the peptide, and the potential utility in other systems.

Weaknesses:
-Peptide labeling specificity and efficiency is not clear. High levels of background labeling. While it was sufficient for demonstrating the system works, it may pose problems depending on the peptide sequence, and/or use at lower dose.
-Only one peptide system was tested, namely OVA323-339 region.

-Limited novel biological findings. This study mostly describes a new tool that may have exciting potential.

- https://doi.org/10.7554/eLife.91809.2.sa0

Author Response

The following is the authors' response to the original reviews.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *The authors develop a method to fluorescently tag peptides loaded onto dendritic cells using a two-step method with a tetracystein motif modified peptide and labelling step done on the surface of live DC using a dye with high affinity for the added motif. The results are convincing in demonstrating in vitro and in vivo T cell activation and efficient label transfer to specific T cells in vivo. The label transfer technique will be useful to identify T cells that have recognised a DC presenting a specific peptide antigen to allow the isolation of the T cell and cloning of its TCR subunits, for example. It may also be useful as a general assay for in vitro or in vivo T-DC communication that can allow the detection of genetic or chemical modulators.*
>
> *Strengths:*
>
> *The study includes both in vitro and in vivo analysis including flow cytometry and two-photon laser scanning microscopy. The results are convincing and the level of T cell labelling with the fluorescent pMHC is surprisingly robust and suggests that the approach is potentially revealing something about fundamental mechanisms beyond the state of the art.*
>
> *Weaknesses:*
>
> *The method is demonstrated only at high pMHC density and it is not clear if it can operate at at lower peptide doses where T cells normally operate. However, this doesn't limit the utility of the method for applications where the peptide of interest is known. It's not clear to me how it could be used to de-orphan known TCR and this should be explained if they want to claim this as an application. Previous methods based on biotin-streptavidin and phycoerythrin had single pMHC sensitivity, but there were limitations to the PE-based probe so the use of organic dyes could offer advantages.*

We thank the reviewer for the valuable comments and suggestions. Indeed, we have shown and optimized this labeling technique for a commonly used peptide at rather high doses to provide a proof of principle for the possible use of tetracysteine tagged peptides for in vitro and in vivo studies. However, we completely agree that the studies that require different peptides and/or lower pMHC concentrations may require preliminary experiments if the use of biarsenical probes is attempted. We think it can help investigate the functional and biological properties of the peptides for TCRs deorphaned by techniques. Tetracysteine tagging of such peptides would provide a readily available antigen-specific reagent for the downstream assays and validation. Other possible uses for modified immunogenic peptides could be visualizing the dynamics of neoantigen vaccines or peptide delivery methods in vivo. For these additional uses, we recommend further optimization based on the needs of the prospective assay.

> ***Reviewer #2 (Public Review):***
>
> *Summary:*
>
> *The authors here develop a novel Ovalbumin model peptide that can be labeled with a site-specific FlAsH dye to track agonist peptides both in vitro and in vivo. The utility of this tool could allow better tracking of activated polyclonal T cells particularly in novel systems. The authors have provided solid evidence that peptides are functional, capable of activating OTII T cells, and that these peptides can undergo trogocytosis by cognate T cells only.*
>
> *Strengths:*
>
> *-An array of in vitro and in vivo studies are used to assess peptide functionality.*
>
> *-Nice use of cutting-edge intravital imaging.*
>
> *-Internal controls such as non-cogate T cells to improve the robustness of the results (such as Fig 5A-D).*
>
> *-One of the strengths is the direct labeling of the peptide and the potential utility in other systems.*
>
> *Weaknesses:*
>
> 1. *What is the background signal from FlAsH? The baselines for Figure 1 flow plots are all quite different. Hard to follow. What does the background signal look like without FLASH (how much fluorescence shift is unlabeled cells to No antigen+FLASH?). How much of the FlAsH in cells is actually conjugated to the peptide? In Figure 2E, it doesn't look like it's very specific to pMHC complexes. Maybe you could double-stain with Ab for MHCII. Figure 4e suggests there is no background without MHCII but I'm not fully convinced. Potentially some MassSpec for FLASH-containing peptides.*

We thank the reviewer for pointing out a possible area of confusion. In fact, we have done extensive characterization of the background and found that it has varied with the batch of FlAsH, TCEP, cytometer and also due to the oxidation prone nature of the reagents. Because Figure 1 subfigures have been derived from different experiments, a combination of the factors above have likely contributed to the inconsistent background. To display the background more objectively, we have now added the No antigen+Flash background to the revised Fig 1.

It is also worthwhile noting that nonspecific Flash incorporation can be toxic at increasing doses, and live cells that display high backgrounds may undergo early apoptotic changes in vitro. However, when these cells are adoptively transferred and tracked in vivo, the compromised cells with high background possibly undergo apoptosis and get cleared by macrophages in the lymph node. The lack of clearance in vitro further contributes to different backgrounds between in vitro and in vivo, which we think is also a possible cause for the inconsistent backgrounds throughout the manuscript. Altogether, comparison of absolute signal intensities from different experiments would be misleading and the relative differences within each experiment should be relied upon. We have added further discussion about this issue.

> 1. *On the flip side, how much of the variant peptides are getting conjugated in cells? I'd like to see some quantification (HPLC or MassSpec). If it's ~10% of peptides that get labeled, this could explain the low shifts in fluorescence and the similar T cell activation to native peptides if FlasH has any deleterious effects on TCR recognition. But if it's a high rate of labeling, then it adds confidence to this system.*

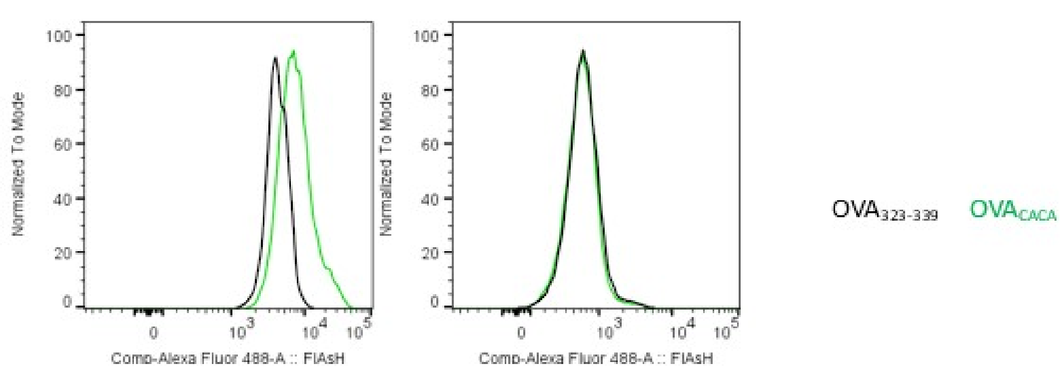
We agree that mass spectrometry or, more specifically tandem MS/MS, would be an excellent addition to support our claim about peptide labeling by FlAsH being reliable and non-disruptive. Therefore, we have recently undertaken a tandem MS/MS quantitation project with our collaborators. However, this would require significant time to determine the internal standard based calibration curves and to run both analytical and biological replicates. Hence, we have decided pursuing this as a follow up study and added further discussion on quantification of the FlAsH-peptide conjugates by tandem MS/MS.

> 1. *Conceptually, what is the value of labeling peptides after loading with DCs? Why not preconjugate peptides with dye, before loading, so you have a cleaner, potentially higher fluorescence signal? If there is a potential utility, I do not see it being well exploited in this paper. There are some hints in the discussion of additional use cases, but it was not clear exactly how they would work. One mention was that the dye could be added in real-time in vivo to label complexes, but I believe this was not done here. Is that feasible to show?*

We have already addressed preconjugation as a possible avenue for labeling peptides. In our hands, preconjugation resulted in low FlAsH intensity overall in both the control and tetracysteine labeled peptides (Author response image 1). While we don't have a satisfactory answer as to why the signal was blunted due to preconjugation, it could be that the tetracysteine tagged peptides attract biarsenical compounds better intracellularly. It may be due to the redox potential of the intracellular environment that limits disulfide bond formation. (PMID: 18159092)

Author response image 1.

Preconjugation yields poor FlAsH signal. Splenic DCs were pulsed with peptide then treated with FlAsH or incubated with peptide-FlAsH preconjugates. Overlaid histograms show the FlAsH intensities on DCs following the two-step labeling (left) and preconjugation (right). Data are representative of two independent experiments, each performed with three biological replicates.



> 1. *Figure 5D-F the imaging data isn't fully convincing. For example, in 5F and 2G, the speeds for T cells with no Ag should be much higher (10-15micron/min or 0.16-0.25micron/sec). The fact that yours are much lower speeds suggests technical or biological issues, that might need to be acknowledged or use other readouts like the flow cytometry.*

We thank the reviewer for drawing attention to this technical point. We would like to point out that the imaging data in fig 5 d-f was obtained from agarose embedded live lymph node sections. Briefly, the lymph nodes were removed, suspended in 2% low melting temp agarose in DMEM and cut into 200 µm sections with a vibrating microtome. Prior to imaging, tissue sections were incubated in complete RPMI medium at 37 °C for 2 h to resume cell mobility. Thus, we think the cells resuming their typical speeds ex vivo may account for slightly reduced T cell speeds overall, for both control and antigen-specific T cells (PMID: 32427565, PMID: 25083865). We have added text to prevent the ambiguity about the technique for dynamic imaging. The speeds in Figure 2g come from live imaging of DC-T cell cocultures, in which the basal cell movement could be hampered by the cell density. Additionally, glass bottom dishes have been coated with Fibronectin to facilitate DC adhesion, which may be responsible for the lower average speeds of the T cells in vitro.

> ***Reviewer #1 (Recommendations For The Authors):***
>
> *Does the reaction of ReAsH with reactive sites on the surface of DC alter them functionally? Functions have been attributed to redox chemistry at the cell surface- could this alter this chemistry?*

We thank the reviewer for the insight. It is possible that the nonspecific binding of biarsenical compounds to cysteine residues, which we refer to as background throughout the manuscript, contribute to some alterations. One possible way biarsenicals affect the redox events in DCs can be via reducing glutathione levels (PMID: 32802886). Glutathione depletion is known to impair DC maturation and antigen presentation (PMID: 20733204). To avoid toxicity, we have carried out a stringent titration to optimize ReAsH and FlAsH concentrations for labeling and conducted experiments using doses that did not cause overt toxicity or altered DC function.

> *Have the authors compared this to a straightforward approach where the peptide is just labelled with a similar dye and incubated with the cell to load pMHC using the MHC knockout to assess specificity? Why is this that involves exposing the DC to a high concentration of TCEP, better than just labelling the peptide? The Davis lab also arrived at a two-step method with biotinylated peptide and streptavidin-PE, but I still wonder if this was really necessary as the sensitivity will always come down to the ability to wash out the reagents that are not associated with the MHC.*

We agree with the reviewer that small undisruptive fluorochrome labeled peptide alternatives would greatly improve the workflow and signal to noise ratio. In fact, we have been actively searching for such alternatives since we have started working on the tetracysteine containing peptides. So far, we have tried commercially available FITC and TAMRA conjugated OVA323-339 for loading the DCs, however failed to elicit any discernible signal. We also have an ongoing study where we have been producing and testing various in-house modified OVA323-339 that contain fluorogenic properties. Unfortunately, at this moment, the ones that provided us with a crisp, bright signal for loading revealed that they have also incorporated to DC membrane in a nonspecific fashion and have been taken up by non-cognate T cells from double antigen-loaded DCs. We are actively pursuing this area of investigation and developing better optimized peptides with low/non-significant membrane incorporation.

Lastly, we would like to point out that tetracysteine tags are visible by transmission electron microscopy without FlAsH treatment. Thus, this application could add a new dimension for addressing questions about the antigen/pMHCII loading compartments in future studies. We have now added more in-depth discussion about the setbacks and advantages of using tetracysteine labeled peptides in immune system studies.

> *The peptide dosing at 5 µM is high compared to the likely sensitivity of the T cells. It would be helpful to titrate the system down to the EC50 for the peptide, which may be nM, and determine if the specific fluorescence signal can still be detected in the optimal conditions. This will not likely be useful in vivo, but it will be helpful to see if the labelling procedure would impact T cell responses when antigen is limited, which will be more of a test. At 5 µM it's likely the system is at a plateau and even a 10-fold reduction in potency might not impact the T cell response, but it would shift the EC50.*

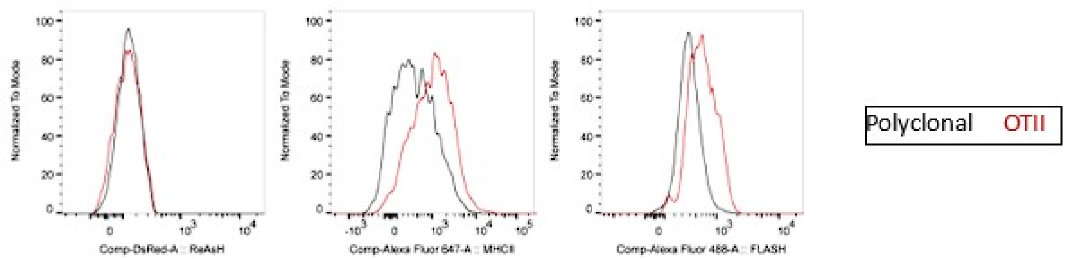
We thank the reviewer for the comment and suggestion. We agree that it is possible to miss minimally disruptive effects at 5 µM and titrating the native peptide vs. modified peptide down to the nM doses would provide us a clearer view. This can certainly be addressed in future studies and also with other peptides with different affinity profiles. A reason why we have chosen a relatively high dose for this study was that lowering the peptide dose had costed us the specific FlAsH signal, thus we have proceeded with the lowest possible peptide concentration.

> *In Fig 3b the level of background in the dsRed channel is very high after DC transfer. What cells is this associated with and does this appear be to debris? Also, I wonder where the ReAsH signal is in the experiments in general. I believe this is a red dye and it would likely be quite bright given the reduction of the FlAsH signal. Will this signal overlap with signals like dsRed and PHK-26 if the DC is also treated with this to reduce the FlAsH background?*

We have already shown that ReAsH signal with DsRed can be used for cell-tracking purposes as they don't get transferred to other cells during antigen specific interactions (Author response image 2). In fact, combining their exceptionally bright fluorescence provided us a robust signal to track the adoptively transferred DCs in the recipient mice. On the other hand, the lipophilic membrane dye PKH-26 gets transferred by trogocytosis while the remaining signal contributes to the red fluorescence for tracking DCs. Therefore, the signal that we show to be transferred from DCs to T cells only come from the lipophilic dye. To address this, we have added a sentence to elaborate on this in the results section. Regarding the reviewer's comment on DsRed background in Figure 3b., we agree that the cells outside the gate in recipient mice seems slightly higher that of the control mice. It may suggest that the macrophages clearing up debris from apoptotic/dying DCs might contribute to the background elicited from the recipient lymph node. Nevertheless, it does not contribute to any DsRed/ReAsH signal in the antigen-specific T cells.

Author response image 2.

ReAsH and DsRed are not picked up by T cells during immune synapse. DsRed+ DCs were labeled with ReAsH, pulsed with 5 µM OVACACA, labeled with FlAsH and adoptively transferred into CD45.1 congenic mice mice (1-2 × 106 cells) via footpad. Naïve e450-labeled OTII and e670-labeled polyclonal CD4+ T cells were mixed 1:1 (0.25-0.5 × 106/ T cell type) and injected i.v. Popliteal lymph nodes were removed at 42 h post-transfer and analyzed by flow cytometry. Overlaid histograms show the ReAsh/DsRed, MHCII and FlAsH intensities of the T cells. Data are representative of two independent experiments with n=2 mice per group.

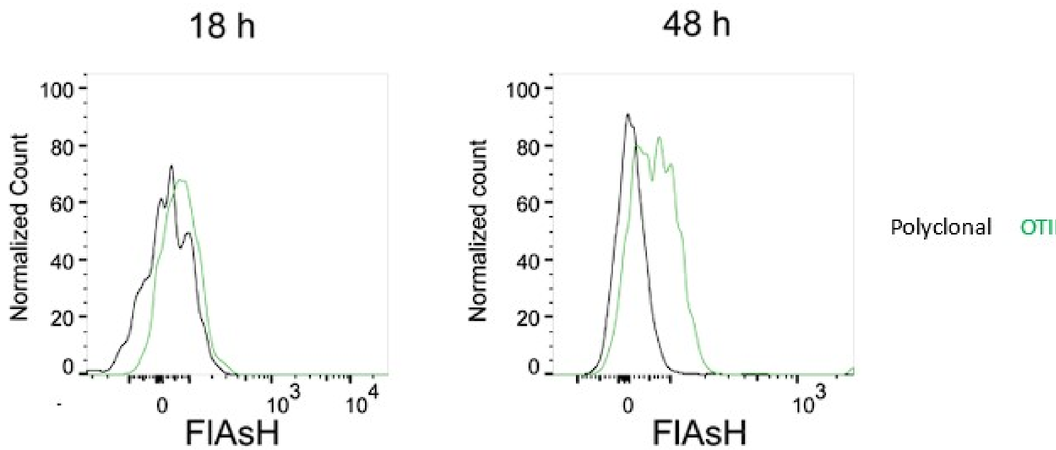


> *In Fig 5b there is a missing condition. If they look at Ea-specific T cells for DC with without the Ova peptide do they see no transfer of PKH-26 to the OTII T cells? Also, the FMI of the FlAsH signal transferred to the T cells seems very high compared to other experiments. Can the author estimate the number of peptides transferred (this should be possible) and would each T cell need to be collecting antigens from multiple DC? Could the debris from dead DC also contribute to this if picked up by other DC or even directly by the T cells? Maybe this could be tested by transferring DC that are killed (perhaps by sonication) prior to inoculation?*

To address the reviewer's question on the PKH-26 acquisition by T cells, Ea-T cells pick up PKH-26 from Ea+OVA double pulsed DCs, but not from the unpulsed or single OVA pulsed DCs. OTII T cells acquire PKH-26 from OVA-pulsed DCs, whereas Ea T cells don't (as expected) and serve as an internal negative control for that condition. Regarding the reviewer's comment on the high FlAsH signal intensity of T cells in Figure 5b, a plausible explanation can be that the T cells accumulate pMHCII through serial engagements with APCs. In fact, a comparison of the T cell FlAsH intensities 18 h and 36-48 h post-transfer demonstrate an increase (Author response image 3) and thus hints at a cumulative signal. As DCs are known to be short-lived after adoptive transfer, the debris of dying DCs along with its peptide content may indeed be passed onto macrophages, neighboring DCs and eventually back to T cells again (or for the first time, depending on the T:DC ratio that may not allow all T cells to contact with the transferred DCs within the limited time frame). We agree that the number and the quality of such contacts can be gauged using fluorescent peptides. However, we think peptides chemically conjugated to fluorochromes with optimized signal to noise profiles and with less oxidation prone nature would be more suitable for quantification purposes.

Author response image 3.

FlAsH signal acquisition by antigen specific T cells becomes more prominent at 36-48 h post-transfer. DsRed+ splenic DCs were double-pulsed with 5 μM OVACACA and 5 μM OVA-biotin and adoptively transferred into CD45.1 recipients (2 × 106 cells) via footpad. Naïve e450-labeled OTII (1 × 106 cells) and e670-labeled polyclonal T cells (1 × 106 cells) were injected i.v. Popliteal lymph nodes were analyzed by flow cytometry at 18 h or 48 h post-transfer. Overlaid histograms show the T cell levels of OVACACA (FlAsH). Data are representative of three independent experiments with n=3 mice per time point



> ***Reviewer #2 (Recommendations For The Authors):***
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> *As mentioned in weaknesses 1 & 2, more validation of how much of the FlAsH fluorescence is on agonist peptides and how much is non-specific would improve the interpretation of the data. Another option would be to preconjugate peptides but that might be a significant effort to repeat the work.*

We agree that mass spectrometry would be the gold standard technique to measure the percentage of tetracysteine tagged peptide is conjugated to FlAsH in DCs. However, due to the scope of such endevour this can only be addressed as a separate follow up study. As for the preconjugation, we have tried and unfortunately failed to get it to work (Reviewer Figure 1). Therefore, we have shifted our focus to generating in-house peptide probes that are chemically conjugated to stable and bright fluorophore derivates. With that, we aim to circumvent the problems that the two-step FlAsH labeling poses.

> *Along those lines, do you have any way to quantify how many peptides you are detecting based on fluorescence? Being able to quantify the actual number of peptides would push the significance up.*

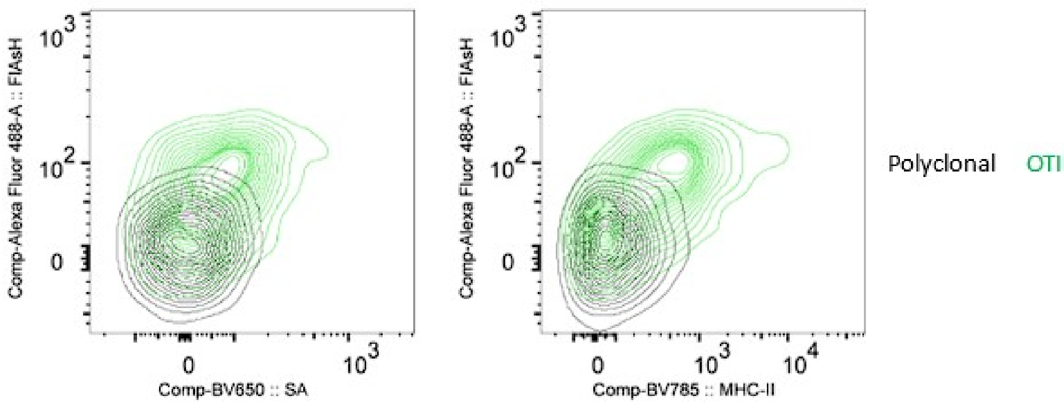
We think two step procedure and background would pose challenges to such quantification in this study. although it would provide tremendous insight on the antigen-specific T cell- APC interactions in vivo, we think it should be performed using peptides chemically conjugated to fluorochromes with optimized signal to noise profiles.

> *In Figure 3D or 4 does the SA signal correlate with Flash signal on OT2 cells? Can you correlate Flash uptake with T cell activation, downstream of TCR, to validate peptide transfers?*

To answer the reviewer's question about FlAsH and SA correlation, we have revised the Figure 3d to show the correlation between OTII uptake of FlAsH, Streptavidin and MHCII. We also thank the reviewer for the suggestion on correlating FlAsH uptake with T cell activation and/or downstream of TCR activation. We have used proliferation and CD44 expressions as proxies of activation (Fig 2, 6). Nevertheless, we agree that the early events that correspond to the initiation of T-DC synapse and FlAsH uptake would be valuable to demonstrate the temporal relationship between peptide transfer and activation. Therefore, we have addressed this in the revised discussion.

Author response image 4.

FlAsH signal acquisition by antigen specific T cells is correlates with the OVA-biotin (SA) and MHCII uptake. DsRed+ splenic DCs were double-pulsed with 5 μM OVACACA and 5 μM OVA-biotin and adoptively transferred into CD45.1 recipients (2 × 106 cells) via footpad. Naïve e450-labeled OTII (1 × 106 cells) and e670-labeled polyclonal T cells (1 × 106 cells) were injected i.v. Popliteal lymph nodes were analyzed by flow cytometry. Overlaid histograms show the T cell levels of OVACACA (FlAsH) at 48 h post-transfer. Data are representative of three independent experiments with n=3 mice.



> *Minor:*
>
> *Figure 3F, 5D, and videos: Can you color-code polyclonal T cells a different color than magenta (possibly white or yellow), as they have the same look as the overlay regions of OT2-DC interactions (Blue+red = magenta).*

We apologize for the inconvenience about the color selection. We have had difficulty in assigning colors that are bright and distinct. Unfortunately, yellow and white have also been easily mixed up with the FlAsH signal inside red and blue cells respectively. We have now added yellow and white arrows to better point out the polyclonal vs. antigen specific cells in 3f and 5d.